

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Concept Mapping

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
Student Name:	Murshidha M	Reg. No.:	192571007
Section / Batch:	Slot A	Date:	10/07/2026

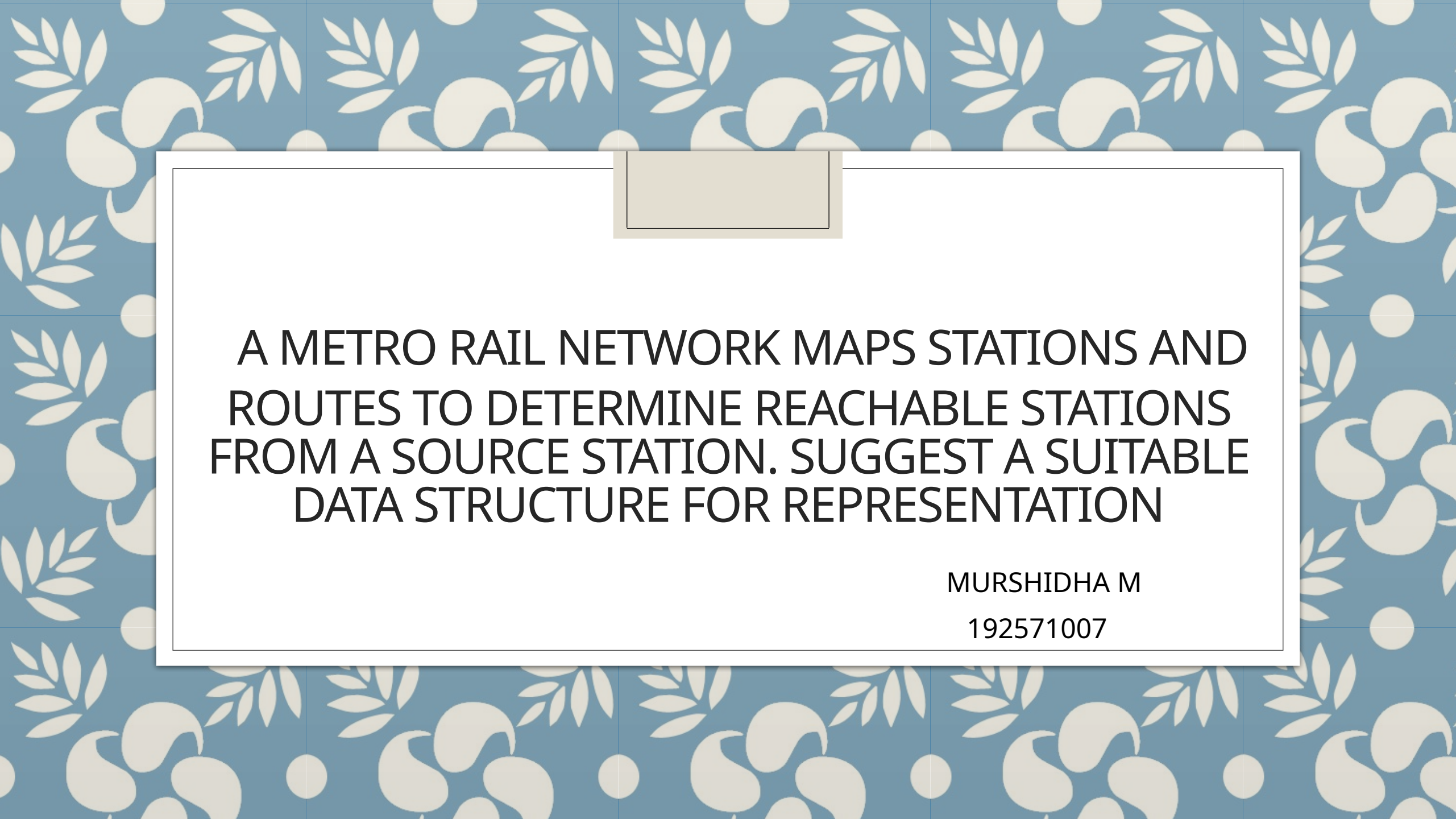
Code of Conduct

I Murshidha M (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: _____

Instructions

- This test contains 5 Concept mapping questions. Total: 100 Marks.
- When a student answer using a concept map, in evaluation the answer should be with following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	2	20
Search Data Structure	Linear Search and Binary Search	3	20
Recursion, Performance analysis	Recursion	4	20
Tree Data Structures	Classifications of Tree data structure	5	20
GRAND TOTAL:			100 Marks



**A METRO RAIL NETWORK MAPS STATIONS AND
ROUTES TO DETERMINE REACHABLE STATIONS
FROM A SOURCE STATION. SUGGEST A SUITABLE
DATA STRUCTURE FOR REPRESENTATION**

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Abstract

- A metro rail network consists of stations connected by railway routes.
- A suitable data structure is needed to represent stations and their connections efficiently.
- **Graph** is the most appropriate data structure for modeling a metro network.
- In a graph, **stations are represented as vertices (nodes)** and **routes are represented as edges**.
- Graphs help determine all stations that are reachable from a given source station.
- Graph traversal algorithms such as **Breadth-First Search (BFS)** and **Depth-First Search (DFS)** are used to find reachable stations.
- This representation supports efficient route planning, connectivity analysis, and network management.
- Graph-based models are widely used in real-world metro navigation and transportation systems.

INTRODUCTION

- A metro rail network contains many stations connected through different routes.
- To store and manage these connections efficiently, an appropriate data structure is required.
- A **Graph** is the best data structure because it naturally represents stations and their connections.
- In a graph:
 - **Vertices (Nodes)** represent metro stations.
 - **Edges** represent railway routes between stations.
- Graph algorithms such as **Breadth-First Search (BFS)** and **Depth-First Search (DFS)** help determine which stations are reachable from a given source station.
- This representation improves route planning, connectivity analysis, and overall network management.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____